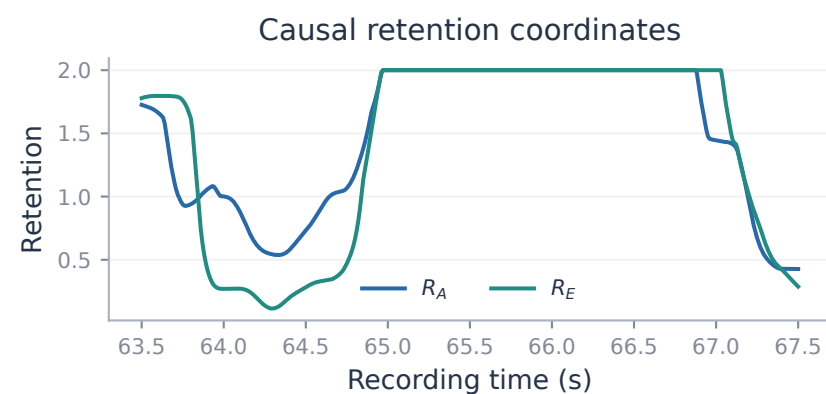
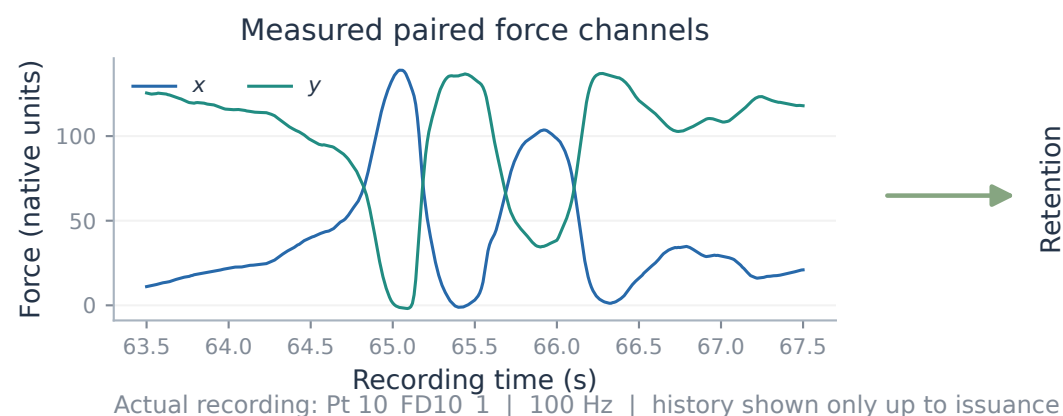


a From paired force signals to an interpretable movement state

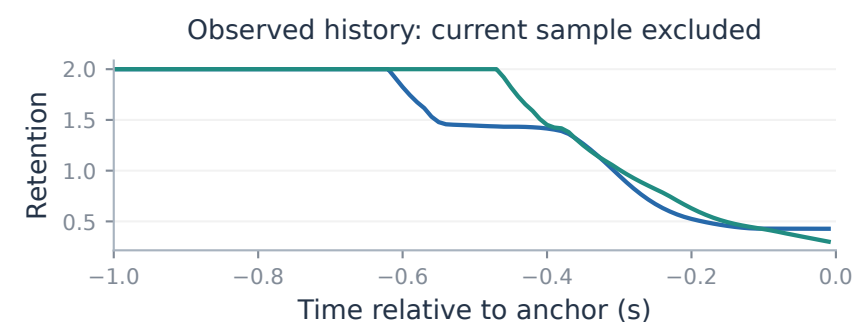


Amplitude-energy state

$$z(t) = [R_A(t), R_E(t)]$$

Oscillatory motivation; no fitted-context input

b Causal history and the separate current-state anchor



100 preceding samples × 4 features

$$\mathcal{H}_t^z = \{R_A, b_A, R_E, b_E\}_{t-1s \leq \tau < t}$$

Backward slopes • frozen feature scaling

Separate anchor

$$z_t = z(t)$$

At $t = 67.500$ s: (0.427, 0.288)

c State-history operator: history branch, horizon trunk and persistence residual

State-history branch

Gate + time weights → 400 → 128 → 62

Forecast-horizon trunk

$(\eta, \eta^2, \eta^3) \rightarrow 64 \rightarrow 62$

$\eta = h/H$; $H = 1.5$ s

Feature fusion

$$V_z \odot V_h$$

Residual head

62 → 64 → 2

Anchored future state

$$\hat{z}_j(t+h) = \Pi_{[0,2]}[z_{t,j} + \sigma_j \eta a_{\theta,j}(\mathcal{H}_t^z, h)]$$

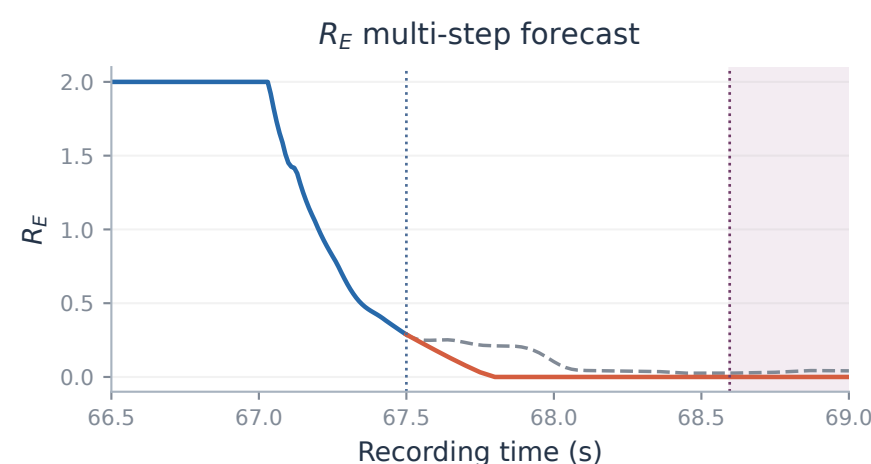
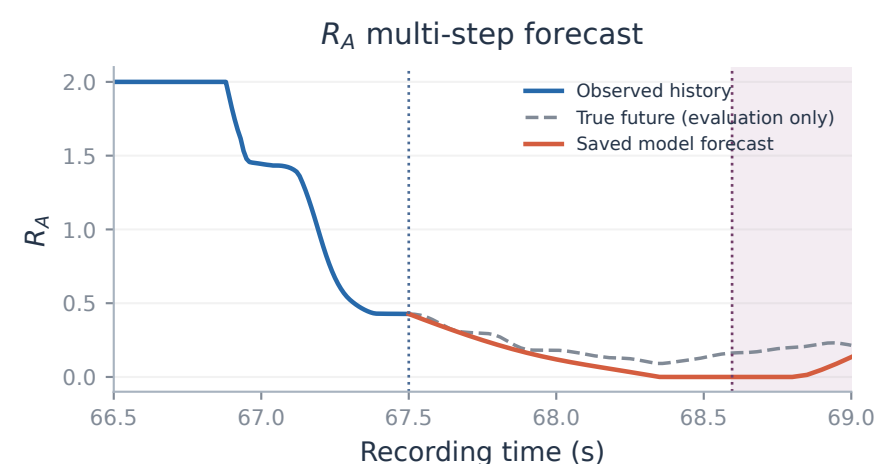
$$\hat{z}(t+0) = z_t$$

σ_j : target scale; $a_{\theta,j}$: head output

30 queried horizons: 0.05–1.50 s

State-history model: 68,282 trainable parameters

d Saved multi-horizon forecasts and an actual issued warning



Frozen trajectory-to-warning rule

Low / declining → 3 future points
→ current support → 3 rolling anchors → issuance
Exit: candidate absent for 0.10 s

Lead = 1.096 s

Issued alarm
 $a = 67.500$ s

Reference onset
 $g = 68.596$ s

Blue: available history Orange: saved forecast Grey dashed: held-out observations Shading: algorithmic reference event

Recorded example: Pt 10_FD10_1 (seed 2026); illustrates the forecasting and warning procedure.